

GIRGUARD AI — PROBLEM STATEMENT

Human-Wildlife Conflict Prediction & Mitigation Platform | Gujarat Forest Department

1. Executive Summary & Context

The Gir Protected Area Network in Gujarat, India, is the sole remaining natural habitat of the endangered Asiatic Lion (*Panthera leo persica*) and home to over 300 Asiatic leopards. With the lion population exceeding 674 individuals and expanding beyond sanctuary boundaries into agricultural human settlements across Junagadh, Amreli, Gir Somnath, and Bhavnagar districts, human-wildlife interactions have escalated sharply. Over 70% of Asiatic lions now range outside the core protected national park, resulting in livestock depredation, accidental human injury, vehicle collisions on railway and road corridors, and retaliatory poisoning risks.

2. The Core Problem Statement

Traditional wildlife monitoring relies heavily on manual ranger foot patrols, reactive phone calls from villagers, and delayed camera trap retrievals. This reactive model suffers from critical operational bottlenecks:

- **High Latency in Warning Dispatch:** Villagers receive warnings hours after carnivores enter agricultural fields, preventing preventive action.
- **Resource Misallocation:** Rapid Response Teams (RRT) are dispatched without real-time spatial telemetry or predictive trajectory data.
- **Delayed Compensation Disbursal:** Crop and livestock damage claims take an average of 30-45 days to verify manually, causing community resentment.
- **Fragmented Telemetry:** GPS collars, thermal camera trap arrays, and waterhole sensor data operate in isolated silos without unified predictive intelligence.

3. The GirGuard AI Solution Architecture

GirGuard AI introduces an enterprise-grade Command & Control Web Application that aggregates live GPS collar telemetry, thermal camera trap feeds, and weather/topography layers into an integrated predictive AI model.

Module / Component	Technology & AI Mechanism	Operational Objective
Live Movement Telemetry	GPS collar spatial tracking + velocity vectoring	Continuous tracking of carnivore movement near village perimeters
Early Warning Dispatch	Multilingual SMS, WhatsApp, & Automated Sirens	Push sub-30 second early warnings to high-risk village volunteers.
Rapid Response Command	Real-time squad dispatch & vehicle ETA tracker	Optimize RRT unit routing to intercept straying lions safely.
AI Compensation Hub	Computer vision damage verification	Accelerate claim approval from weeks to under 48 hours.
Hotspot Analytics	Spatial density & kernel density estimation	Identify seasonal high-conflict corridors for fence placement.

4. Target Impact & Metrics

- **Zero Human Casualties:** Achieve 100% early warning coverage for core fringe villages within 2 km of carnivore movement.
- **Sub-15 Minute Dispatch:** Reduce Rapid Response Team intercept time from 45 mins to under 15 mins.
- **87%+ Model Accuracy:** AI spatial prediction accuracy for carnivore movement trajectory over 6-hour windows.

- **Community Trust:** Direct compensation disbursal cycle reduced to 48 hours, eliminating human retaliation.